Aleksei Andreev

Senior Software Engineer

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Professional Summary

Senior Software Engineer with 15+ years of experience designing and developing high-performance real-time software systems across industrial robotics, simulation, graphics, VR/AR, embedded systems, and IoT domains.

Specialized in computational geometry, CAD processing, topology analysis, GPU-accelerated algorithms, and performance-critical software architecture. Proven ability to solve complex multidisciplinary engineering challenges spanning software, hardware, mathematics, and real-world automation systems.

Experienced in delivering end-to-end solutions ranging from industrial robotics platforms and geometry-processing pipelines to immersive VR applications, custom hardware integrations, embedded firmware, and content-authoring tools. Recognized for strong analytical thinking, rapid learning, and the ability to bridge multiple technical domains to deliver robust and practical engineering solutions.

Technical Expertise

Programming Languages

- C#
- C++
- Python
- JavaScript / TypeScript
- GLSL
- HLSL

Core Engineering Domains

- Real-Time Systems
- Computational Geometry
- Simulation Systems
- Graphics Programming
- Software Architecture
- Performance Optimization
- Mathematical Modeling
- GPU Computing

Robotics & Geometry Processing

- Industrial Robotics
- CAD Processing
- STEP
- OpenCascade
- Topological Analysis
- Feature Extraction
- Path Planning Workflows
- 3D Geometry Processing

XR & Interactive Applications

- Unity
- VR
- AR
- Procedural Generation
- Shader Development
- Physics Simulation
- NVIDIA PhysX

Embedded & IoT

- Firmware Development
- Nordic nRF Connect SDK
- Zephyr RTOS
- Bluetooth Low Energy (BLE)
- GATT
- Cellular Connectivity
- Hardware Integration

Selected Technical Achievement

High-Performance CAD Topology Analysis System

Designed and implemented a custom topology analysis system for CAD models to support automated feature extraction and robotic manufacturing workflows.

Developed specialized computational geometry algorithms optimized for industrial-scale datasets after evaluating existing solutions and determining that available approaches did not meet performance requirements.

The resulting system enabled efficient extraction of manufacturing-relevant geometric features and became a key component of the geometry-processing pipeline used for robotic trajectory generation.

PROFESSIONAL EXPERIENCE

Augmentus Robotics: Senior Software Developer

August 2025 - February 2026, Singapore

Contributed to the development of an industrial robotics programming platform focused on manufacturing automation and robotic path-planning workflows.

- Designed and implemented an OpenCascade-based CAD processing pipeline supporting STEP model cleanup, topology filtering, feature selection, and geometry preparation for trajectory generation.
- Developed a high-performance topology analysis system for automated feature extraction from CAD models using custom computational geometry algorithms.
- Implemented software integrity protection mechanisms to detect and prevent unauthorized modification of deployed applications.
- Integrated and extended third-party libraries while contributing to core platform functionality and system maintenance.
- Collaborated across robotics, geometry-processing, and application-development domains to support industrial automation workflows.

LAI Games: Senior Game Programmer

November 2021 - August 2024, Singapore

Developed software solutions spanning VR applications, custom hardware, firmware, and middleware systems.

- Developed gameplay systems and interactive experiences for VR applications with a focus on immersion, responsiveness, and performance.
- Designed and implemented virtual arcade machine systems for the Arcade Legends VR platform.
- Developed firmware for custom arcade cabinet hardware, enabling communication between physical devices and VR software.
- Built middleware and communication services supporting reliable hardware-software integration.
- Worked across software and hardware layers to deliver integrated entertainment systems combining VR, embedded devices, and custom peripherals.

KWEST: Lead Software Engineer

December 2016 - May 2021, Dallas, Texas, United States

Played a key role in the development of Kwest Go, a location-based augmented reality storytelling platform combining geospatial gameplay and interactive narratives.

- Architected and implemented core application systems using Unity and Android technologies.
- Designed and developed Kwest Graph, a visual content-authoring framework enabling non-programmers to create interactive experiences without engineering support.
- Developed geolocation and mapping systems supporting location-aware gameplay and narrative content.
- Contributed to the migration of AR functionality from Kudan technologies to Unity XR.
- Participated in technical decision-making, hiring activities, production workflow design, quality assurance, and platform evolution within a distributed international team.

Supervixi Automata: **Independent Developer**

August 2015 - December 2016, Singapore

Developed experimental software products and independent game projects while exploring advanced simulation and content-generation technologies.

- Designed and implemented gameplay systems, simulation mechanics, and procedural content-generation solutions.
- Built GPU-accelerated processing services and automated content-production pipelines.
- Developed full-stack software prototypes exploring AI-assisted content generation and distributed systems.
- Researched and prototyped innovative approaches to interactive simulation and emergent gameplay systems.

Suricate Games: **Software Developer**

September 2012 - June 2015, St Petersburg, Russian Federation

Contributed to the development of simulation and action games using C++ and the Unigine engine.

- Developed gameplay systems and game mechanics for large-scale commercial projects.
- Implemented new features, resolved defects, and contributed to overall game stability and performance.
- Worked within multidisciplinary teams of designers, artists, and engineers to deliver production-quality software.
- Gained extensive experience with large codebases, real-time systems, and collaborative software development processes.

Independent Projects & Technical Interests

Unity Asset Store Publisher

Published tools and extensions for the Unity ecosystem, focusing on developer productivity and technical workflows.

Embedded & IoT Development

Develop embedded and connected-device solutions using Nordic Semiconductor platforms, Zephyr RTOS, BLE GATT services, and cellular connectivity.

Simulation-Driven Game Development

Currently developing a survival-focused card game featuring deep simulation systems, procedural content generation, emergent gameplay, and complex interactions inspired by sandbox survival and colony-simulation games.

Research Interests

Regularly explore computational geometry, rendering techniques, simulation systems, procedural generation, optimization, and performance-critical software development through personal projects and technical research.